

vulnerabilities slipping through the cracks.

Collaboration across teams: In a traditional development process, development, security, and operations teams often work in silos. DevSecOps encourages collaboration among these teams, fostering a culture where everyone is responsible for security.

There are a variety of open source tools that help automate security processes within the DevOps pipeline, from configuration management and code analysis to vulnerability scanning and incident management.

- **Configuration management tools:** Puppet, Chef, and Ansible
- **Continuous integration and continuous deployment (CI/CD) tools:** Jenkins, GitLab, and CircleCI
- **Security information and event management (SIEM) tools:** ELK Stack and Graylog

Building a secure and scalable cloud infrastructure

As organisations move towards cloud-based infrastructure, managing and securing cloud environments becomes a critical task. Cloud environments offer scalability, flexibility, and cost-efficiency, but they also introduce new challenges related to security, compliance, and resource management. Open source tools play a crucial role in building and maintaining secure and scalable cloud infrastructures, enabling organisations to automate deployment, manage resources, and enforce security policies at scale.

OpenStack: OpenStack is an open source cloud computing platform that enables organisations to deploy and manage private clouds. It provides a suite of services for computing, networking, storage, and identity management, making it an ideal choice for organisations that want to create and manage their own cloud infrastructure.

Its key features are:

- **Compute (Nova):** OpenStack Nova manages virtual machines (VMs)

and instances in the cloud, providing scalable compute resources.

- **Storage (Cinder and Swift):** OpenStack provides block storage (Cinder) and object storage (Swift) for managing cloud data.
- **Networking (Neutron):** Neutron manages networking services, including IP address management, DNS, and load balancing.
- **Identity and access management (Keystone):** OpenStack Keystone provides authentication and authorisation services, ensuring that users have secure access to cloud resources.

Kubernetes: Kubernetes is an open source platform for automating the deployment, scaling, and management of containerised applications. Kubernetes abstracts the underlying infrastructure and allows organisations to run applications on any cloud platform, whether it's a private cloud, a public cloud, or a hybrid environment. It has become the *de facto* standard for container orchestration and plays a crucial role in managing modern cloud-native applications. Key features include:

- **Container orchestration:** Kubernetes manages the lifecycle of containerised applications, automating tasks like deployment, scaling, and load balancing.
- **Self-healing:** Kubernetes automatically restarts failed containers, replaces containers, and reschedules them to healthy nodes to ensure high availability.
- **Automated scaling:** Kubernetes can scale applications up or down automatically based on demand, ensuring that cloud resources are used efficiently.

Terraform: Terraform is an open source Infrastructure as Code (IaC) tool that allows users to define and provision cloud infrastructure through code. Terraform is cloud-agnostic and can be used to manage resources across various cloud providers, including AWS, Azure, Google Cloud, and

OpenStack. With Terraform, developers and infrastructure teams can define their cloud infrastructure using declarative configuration files, allowing them to automate provisioning, updates, and scaling. Its key features are:

- **Infrastructure as Code (IaC):** Terraform allows you to write infrastructure configurations as code, making it easier to version, test, and share infrastructure definitions.
- **Multi-cloud support:** Terraform supports multiple cloud providers, enabling organisations to manage hybrid and multi-cloud environments from a single codebase.
- **State management:** Terraform tracks the state of the infrastructure, ensuring that the actual infrastructure matches the desired state defined in the configuration files.

Simplifying network configuration and management with network automation

As IT infrastructures grow in size and complexity, network automation has become a key factor in managing and securing networks efficiently. Manual configuration of network devices and services is time-consuming and prone to errors that can lead to security vulnerabilities and operational inefficiencies. Open source tools like Ansible and NetBox are revolutionising the way network configurations are managed by enabling automation, ensuring consistency, and reducing the possibility of human error.

Ansible: Ansible is an open source automation tool that simplifies configuration management and automation of IT infrastructure. It uses a simple, YAML-based language (called Ansible Playbooks) to define automation tasks and configurations, making it easy for network administrators to automate a wide range of networking tasks.

Its key features for network automation are: